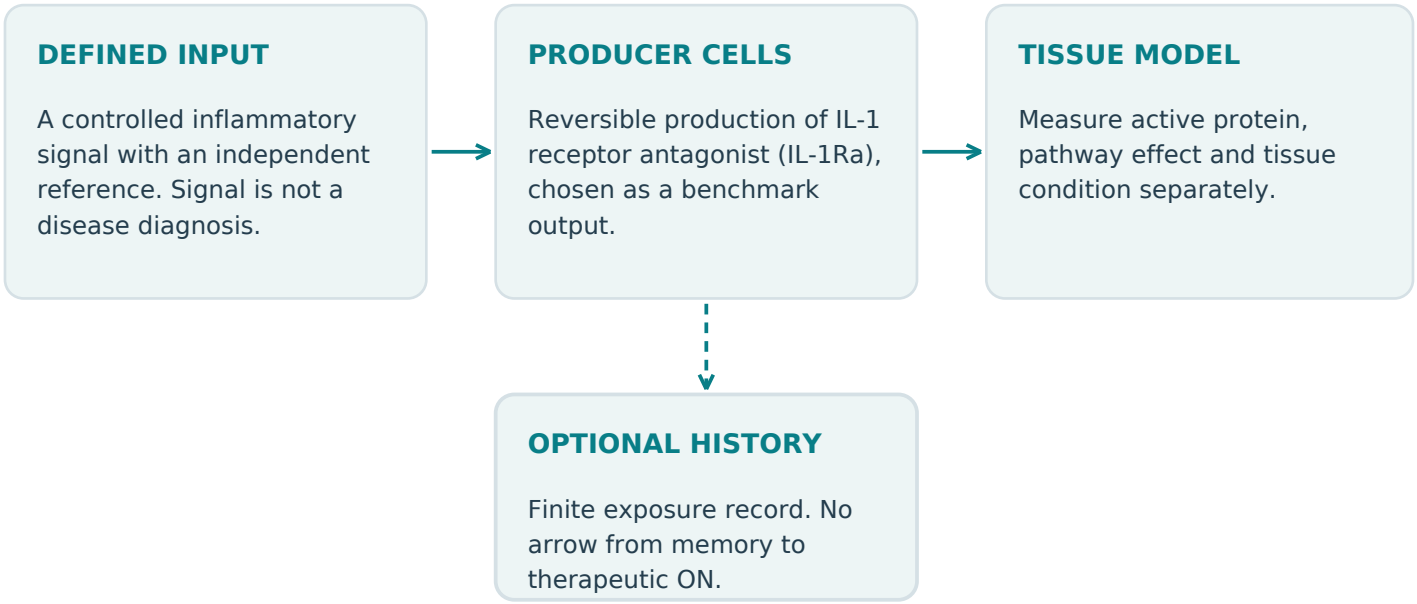
A THERAPEUTIC FORK OF ECHO-M

Artificial medicine, made by cells

A responsive biologic-production platform: one defined signal, one therapeutic protein, measurable exposure and separate evidence for stopping.

Design choice: develop the therapeutic function first; qualify the artificial chromosome as a separate carrier branch. The initial scope is a contained cultured-cell / tissue-model system. A future implant is a research direction.



Measure input, secretion, residual activity, cells and tissue outcomes independently.

The intended gain

Match production to a changing, specified input while preserving a way to inspect performance and investigate failure. Any advantage over administered protein or a simpler engineered-cell circuit must be demonstrated.

What changes from ECHO-M

The live branch becomes a producer, with a pharmacology and exposure problem. Historical DNA remains an optional research record. Chromosome persistence, useful memory and therapeutic benefit become three different questions.

Status: concept plus a checked mathematical illustration. No ECHO-Rx cells, therapeutic experiments, animal results or patient results exist.

FUNCTIONAL ARCHITECTURE

Give every module its own evidence

The proposed system separates molecular behavior from the external labels used to describe it.

BLOCK	PROPOSED FUNCTION	REQUIRED DISTINCTION
Sensor	Respond to one declared inflammatory input in a chosen mammalian-cell background.	Sensitivity, specificity, delay and cross-talk; a falling signal can reflect inhibition rather than recovery.
Producer	Make one biologically active protein with reversible input-linked regulation.	Measure active output, basal leak, residual synthesis and cell-to-cell variability. A reporter is not the drug.
External control	Qualify an actual physical or biochemical enable/disable intervention.	A human approval, software flag or ledger entry has no molecular force. Mechanism and escape rate remain open requirements.
Carrier branch	Compare a simpler non-HAC cassette with a separately qualified human artificial chromosome (HAC).	Specify integration behavior for each carrier. HAC location, structure, copy state and function need separate assays.
Optional archive	Retain the original finite exposure-history idea for later research.	A one-event bit saturates. It gives no dose history, timestamps or proof of successful treatment, and never latches production ON.
Containment / ledger	Account for producer cells and link measurements to their specimen, clone, cohort and window.	Containment, viable-cell escape, signal competence, active protein and tissue recovery cannot certify one another.

First benchmark: IL-1Ra production in a contained inflammatory tissue model. The choice is supported by engineered-cell precedents [1-3]; it is not a recommendation for a patient, a universal arthritis treatment or a claim that ECHO-Rx is effective.

Carrier selection and cell identity are unresolved. No intended host-genome editing is retained for the HAC branch; published host-genome-engineered comparator systems do not establish that property for ECHO-Rx.

MASS BALANCE / PROPOSED MODEL

Track the protein before and after release

A bookkeeping model for hypothesis testing. Every rate, compartment and observation link needs experimental qualification.

N: viable producer-cell count. **B**: total active intracellular protein mass across the tracked producer population. **A_D, A_T, A_S**: active mass in depot, target and spillover pools. Rates k, g and d_N have units 1/time; r_syn has mass/cell/time; concentrations are C_j = A_j / V_j.

```
dN/dt   = (g - d_N) N

dB/dt   = N r_syn - (k_sec + k_intra + d_N) B

dA_D/dt = k_sec B + phi d_N B - (k_out + k_deg) A_D

dA_T/dt = k_out A_D - (k_clear + k_spill) A_T

dA_S/dt = k_spill A_T - k_sys A_S
```

Here 0 <= phi <= 1 is the fraction of active intracellular mass released on death. The remainder, (1 - phi)d_N B, goes to an explicit inactivation/debris sink. Other clearance terms are tracked as exits from the modeled active pools. Secretion is counted once, through k_sec B.

The control input is a hypothesis, not a guarantee

```
r_syn = r_leak + (r_max - r_leak) L(t) F(I_D(t-tau))
```

L is the effective, measured biochemical enabling state, not a digital permission. F maps sensed input to a bounded response in the model. Neither its shape nor an implementation has been selected. Separate sensing, expression, maturation and secretion delays if the data require them.

Where this model can break

Uniform death content and instantaneous release are assumptions; dying cells may carry different stores or release them slowly. N must include proliferation, death, escape and functional conversion if relevant. Competence loss may require additional producer states. Binding, transport backflow, changing volumes and nonlinear clearance can invalidate this compartment layout.

No dose cap follows from a per-cell cap. N, basal leak, carrier copy state, transport and activity all matter. The linear model is neither patient pharmacokinetics nor an automatically conservative safety bound.

Independent activity, cell-count and compartment measurements are needed. A single fluorescence or medium measurement cannot identify every hidden rate.

A CHECKED ILLUSTRATIVE RESULT

Production OFF can precede the peak

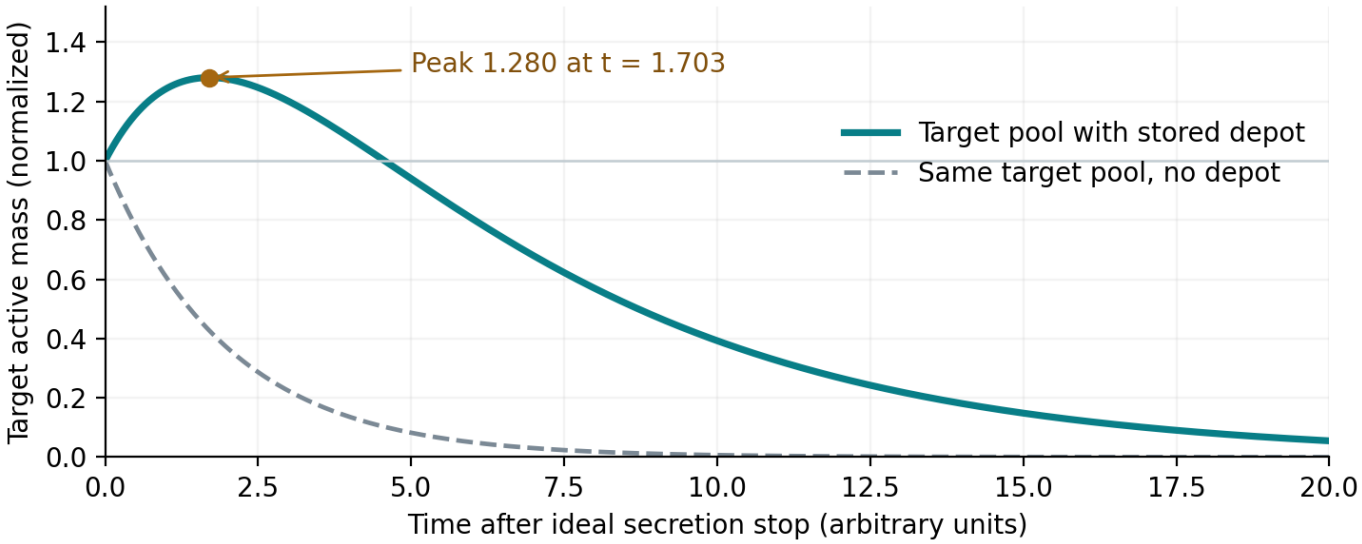
A reservoir can keep feeding the target pool after an idealized, instantaneous cessation of all further secretion.

For the reduced depot/target model, let a = k_out + k_deg, b = k_out and c = k_clear + k_spill. Assume fixed coefficients, no new input, no basal leak and no intracellular or death-release contribution after t = 0.

```
A_T(t) = A_T0 exp(-ct)

       + b A_D0 [exp(-at) - exp(-ct)] / (c-a)

If a=c: A_T(t) = [A_T0 + b A_D0 t] exp(-ct)
```



Arbitrary illustrative values: A_D0 = 6, A_T0 = 1, a = 0.2, b = 0.15, c = 0.5. These have normalized mass and time units; they are not measured biological parameters, a dose or a design target.

```
A_T(t) = 3 exp(-0.2t) - 2 exp(-0.5t)

Peak = 1.28048 at t = 1.70275; post-stop integral = 11.
```

The initial target slope is b A_D0 - c A_T0 = +0.4. A transiently loaded depot produces a peak about 28% above the level at shutdown. At an exact pre-OFF steady state with these fixed coefficients, b A_D0 = c A_T0 and the subsequent target curve instead decreases monotonically.

Verification: analytical solution checked against numerical integration; maximum absolute difference below 10^-8. The equal-rate limit and steady-state monotonicity were also checked. This verifies the example, not the biology.

VERIFICATION ARCHITECTURE

Make the first decision small and testable

Study specifications and observables below are conceptual comparisons, not laboratory procedures or clinical thresholds.

GATE	EVIDENCE TO OBTAIN	DECISION
G0 / Scope	One host background, input, tissue-model endpoint, output, observation window and carrier. Declare thresholds, uncertainty and sampling units before comparison.	No performance label until its decision rule is defined.
G1 / Potency	Independent protein identity/activity, viable producer count and tissue condition. Compare to host-only, inactive-output and externally administered protein references.	Reporter brightness alone cannot support therapeutic function.
G2 / Response	Input-linked secretion and activity under declared absent/present/transient input, unrelated inflammatory cues and repeated cycles.	Separate sensitivity, cross-talk, latency, leak and loss of competence.
G3 / Shutdown	Effective disabling action, intracellular/depot carryover, continuing secretion, residual activity and cell-death effects, assessed independently.	Advance only if the predeclared exposure and function envelope is met; a falling sensor is insufficient.
G4 / Carrier value	Simpler-cassette versus HAC comparison at stated producer counts, exposure window and functional scope, including physical DNA evidence and burden.	Retain HAC only for a documented advantage under explicit boundary requirements; no universal superiority is required.
G5 / Coupling	Requalify the combined carrier/producer/optional archive: common failures, memory burden, clonal shifts, biological drift and missing outcomes.	No transfer of component performance into a system claim without new evidence.

First decisive evidence: a contained system produces independently verified active protein, tracks the declared input, and meets defined leak/shutdown requirements across independent preparations. Failure to meet these criteria stops progression to the HAC/implant claim.

Keep all enrolled units in the accounting, including dead, lost, escaped and unmeasured cells. Report assay linkage, batch/clone dependence, qualification coverage and uncertain outcomes. Replicate wells from one preparation are not automatically independent biological evidence.

FAILURE ANALYSIS

A safe label cannot stop a molecule

The external action is to halt advancement or intervene physically; an UNKNOWN label does not itself alter the cells.

FAILURE	WHY IT MATTERS	REQUIRED EVIDENCE / RESPONSE
Wrong input	Inflammatory pathways can overlap across conditions; inhibition may mask a signal while the underlying problem persists.	Challenge input specificity and use independent tissue-function measures. No infection-safe classifier is claimed.
Basal or stuck output	A healthy-looking reporter may coexist with active therapeutic protein.	Measure secretion and activity independently; qualify the disabling intervention and residual tail.
Producer growth / loss	A per-cell rate bound cannot cap total output; loss can mimic successful OFF.	Measure N, producer competence and stores; account for clonal selection, attrition and copy drift.
Death or lysis	Killing cells may release active stores, debris or other material.	Track release and delayed activity. Do not equate a kill-switch command with clearance.
HAC change	Physical loss, silencing, unintended integration or rearrangement can alter output and history.	Independent location/structure/copy/function evidence, with declared detection limits; no zero-risk inference.
Containment failure	Cell escape, transport limits, oxygen limitation or a host response can undermine a future device.	Qualify each boundary separately before an implant claim. Retrieval does not remove all circulating output or escaped cells.
Blind measurement	A common sensor/reporter failure can make inactivity look like control or recovery.	Independent active-protein and tissue endpoints, specimen linkage and missing-data accounting; classify uncertainty explicitly.

Predeclared stop rules: unexplained persistent activity, breached containment, unacceptable host/tissue burden, or inadequate measurement coverage means HOLD the research claim. Numerical limits remain unset until a model, exposure window and decision are specified.

An implant, if pursued, would need independent qualification of transport, oxygenation, immune response, manufacturing consistency, retrieval and escape detection. Those capabilities are not established by this brief.

COURT DISPOSITION

Critique changed the architecture

Grok, Claude, Gemini and Meta AI reviewed the proposal through the supplied tabs. Their reviews are AI-assisted critique, not independent experimental validation or clinical peer review.

MODEL	ADOPTED	LIMIT OR CORRECTION
Grok	Require a simpler-cassette and administered-protein comparison; qualify output independently of the sensor/reporter.	Simpler systems also have biological failures. Equal performance in one scoped comparison does not refute every possible future HAC use.
Claude	Add total intracellular active mass B, variable producer count N, death release and explicit sinks. Preserve the transient versus steady-state shutdown distinction.	A linear toy model is not a clinical bound. Expected reporter/activity lag is not itself failure. Model mismatch can require a better model rather than refute the platform.
Gemini	Flag overlap between inflammatory sensing and protective responses; require a specificity challenge before therapeutic interpretation.	A marker panel alone cannot certify an infection-safe system. Immune-function or clinical claims need separate evidence. The first reply errored; the shorter review completed.
Meta AI	Qualify a real disabling intervention; separate commanded OFF, secretion, residual activity, cells, DNA and recovery.	An external UNKNOWN/disabled state has no molecular enforcement. Killing or retrieving cells is not instantaneous protein clearance.

Research disposition

Proceed: concept refinement, transparent mathematical stress tests and definition of a contained biological evidence gate.

Pending: cell identity, carrier implementation, molecular enable/disable mechanism, activity assays, useful exposure envelope and all coupled biological measurements.

Not established: new therapeutic efficacy, clinical readiness, reliable human intervention in vivo, superiority, universal programmability or replacement of existing treatment.

Novelty boundary: responsive cell-made biologics already have precedents. The proposed contribution is a falsifiable combination and verification architecture; no first-in-field or patentability claim has been made.

Anubis: concept and design direction. Asteris Solenne / ChatGPT: synthesis, source checks, mathematics and brief. Grok, Claude, Gemini and Meta AI: adversarial review; feedback was checked and selectively adopted.

EVIDENCE AUDIT / NEXT RESEARCH DECISION

What the literature actually supports

Source status checked 28 October 2026. Published component precedents do not validate the ECHO-Rx combination.

[1] Choi et al., 2021. A genome-engineered bioartificial implant for autoregulated anticytokine drug delivery. Science Advances 7, eabj1414.

https://doi.org/10.1126/sciadv.abj1414

Host-genome-engineered cell constructs showed response and efficacy in specific mouse inflammatory-arthritis models. No HAC-based or human ECHO-Rx efficacy is established.

[2] Collins et al., 2023. Hydrogel Encapsulation of Genome-Engineered Stem Cells for Long-Term Self-Regulating Anti-Cytokine Therapy. Gels 9, 169.

https://doi.org/10.3390/gels9020169

Preclinical encapsulated-cell IL-1Ra delivery precedent. Hydrogel encapsulation does not establish the retrieval, immune isolation or escape performance of a new device.

[3] Cimino et al., 2025. Dual-responsive synthetic gene circuit for dynamic biologic drug delivery via inflammatory and circadian signaling pathways. Journal of Biological Engineering 19, 47.

https://doi.org/10.1186/s13036-025-00519-7

Published in-vitro response and tissue-model protection precedent. A circadian module is outside the first ECHO-Rx scope.

[4] Birchak et al., 2025. Rapid assembly of functional modules for generating human artificial chromosome constructs compatible with epigenetic centromere seeding. Chromosome Research 33, 29.

https://doi.org/10.1007/s10577-025-09788-w

Modular HAC assembly and formation precedent in studied cell contexts. This does not establish coupled therapeutic sensing, output control or stable function in the proposed cells.

[5] Birchak et al., 2026. Protein-only centromeric chromatin assembly streamlines human artificial chromosome formation. bioRxiv preprint.

https://doi.org/10.64898/2026.01.28.702397

A research direction for HAC formation. PubMed indexing does not mean peer review. No quantitative formation-efficiency claim is used here.

Next decision to settle

Choose one cultured-cell background, one independently measurable input and one tissue-model endpoint, then specify the comparison and shutdown evidence before selecting a carrier. A negative comparison should be publishable as a result; no forced win for the chromosome is required.

Provenance: therapeutic fork of ECHO-M v0.4 (28 October 2026). That recorder remains a separate research concept. Initial Court briefing miscited the 2021 DOI and 2025 journal; both were corrected in the Court record and in this source list.